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*                                     *
*      STAAD.Pro                     *
*      Version  2005      Bld 1002.US *
*      Proprietary Program of        *
*      Research Engineers, Intl.      *
*      Date=    AUG 17, 2006         *
*      Time=    10:46:27             *
*                                     *
*      USER ID: Meinhardt Singapore Pte Ltd *
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1. STAAD PLANE FRAME 1
INPUT FILE: JA_BLD 2_FR1.STD
2. START JOB INFORMATION
3. JOB NAME - DUBAI METRO PROJECT
4. JOB CLIENT - GOVERNMENT OF DUBAI _ ROADS _TRANSPORT AUTHORITY
5. ENGINEER DATE 17-AUG-06
6. END JOB INFORMATION
7. INPUT WIDTH 79
8. UNIT METER KN
9. JOINT COORDINATES
10. 1 0 6.704 0; 2 1.5 6.704 0; 3 5 6.704 0; 4 14.05 6.704 0; 5 23.1 6.704 0
11. 6 26.6 6.704 0; 7 28.1 6.704 0; 101 0 0 0; 201 28.1 0 0
12. MEMBER INCIDENCES
13. 1 1 2; 2 2 3; 3 3 4; 4 4 5; 5 5 6; 6 6 7; 101 101 1; 201 201 7
14. MEMBER PROPERTY AMERICAN
15. 1 TO 6 101 201 TABLE ST W30X124
16. DEFINE MATERIAL START
17. ISOTROPIC STEEL
18. E 2.05E+008
19. POISSON 0.3
20. DENSITY 76.8195
21. ALPHA 1.2E-005
22. DAMP 0.03
23. END DEFINE MATERIAL
24. CONSTANTS
25. MATERIAL STEEL ALL
26. SUPPORTS
27. 101 201 PINNED
28. LOAD 1 SELFWEIGHT
29. SELFWEIGHT Y -1
30. LOAD 2 SDL
31. JOINT LOAD
32. 1 7 FY -28.5
33. MEMBER LOAD
34. 1 TO 6 UNI GY -6.4
35. LOAD 3 LL
36. JOINT LOAD
37. 1 7 FY -3.4
38. 4 FY -10
39. MEMBER LOAD
40. 1 TO 6 UNI GY -4.8

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FRAME 1

-- PAGE NO. 2

41. 2 CON GY -10 1.4  
42. 2 CMOM GZ -2 1.4  
43. 3 CON GY -10 3.9  
44. 3 CMOM GZ -2 3.9  
45. 4 CON GY -10 2.05  
46. 4 CMOM GZ -2 2.05  
47. 4 CON GY -10 8.05  
48. 4 CMOM GZ -2 8.05  
49. LOAD 4 PH (NOTIONAL HORIZONTAL LOAD)  
50. JOINT LOAD  
51. 1 FX 3.72  
52. LOAD 11 WL1 (INTERNAL SUCTION, WIND ANGLE 0)  
53. MEMBER LOAD  
54. 101 UNI GX 9  
55. 201 UNI GX 2  
56. 1 TO 3 UNI GY 2.3  
57. 4 TO 6 UNI GY -5  
58. LOAD 12 WL2 (INTERNAL PRESSURE, WIND ANGLE 0)  
59. MEMBER LOAD  
60. 101 UNI GX 4.1  
61. 201 UNI GX 7  
62. 1 TO 3 UNI GY 7.3  
63. 4 TO 6 UNI GY 4  
64. LOAD 13 WL3 (INTERNAL SUCTION, WIND ANGLE 90)  
65. MEMBER LOAD  
66. 101 UNI GX -5  
67. 201 UNI GX 5  
68. 1 TO 3 UNI GY -5  
69. 4 TO 6 UNI GY -5  
70. LOAD 14 WL4 (INTERNAL PRESSURE, WIND ANGLE 90)  
71. MEMBER LOAD  
72. 101 UNI GX -9.9  
73. 201 UNI GX 9.9  
74. 1 TO 3 UNI GY 8.9  
75. 4 TO 6 UNI GY 8.9  
76. LOAD 15 WL5 (INTERNAL SUCTION, WIND ANGLE 0, REVERSE WIND)  
77. MEMBER LOAD  
78. 101 UNI GX -2  
79. 201 UNI GX -9  
80. 1 TO 3 UNI GY -5  
81. 4 TO 6 UNI GY 2.3  
82. LOAD 16 WL6 (INTERNAL PRESSURE, WIND ANGLE 0, REVERSE WIND)  
83. MEMBER LOAD  
84. 101 UNI GX -7  
85. 201 UNI GX -4.1  
86. 1 TO 3 UNI GY 4  
87. 4 TO 6 UNI GY 7.3  
88. LOAD 21 EQ (EARTHQUAKE LOAD)  
89. MEMBER LOAD  
90. 101 UNI Y 10.5  
91. 201 UNI Y 10.4  
92. LOAD COMB 1001 1.4DL + 1.6LL  
93. 1 1.4 2 1.4 3 1.6  
94. LOAD COMB 1002 1.4DL + 1.6LL + 1.0PH  
95. 1 1.4 2 1.4 3 1.6 4 1.0  
96. LOAD COMB 1011 1.2DL + 1.2LL + 1.2WL1

FRAME 1

-- PAGE NO. 3

97. 1 1.2 2 1.2 3 1.2 11 1.2  
98. LOAD COMB 1111 1.4DL + 1.4WL1  
99. 1 1.4 2 1.4 11 1.4  
100. LOAD COMB 1211 1.0DL + 1.4WL1  
101. 1 1.0 2 1.0 11 1.4  
102. LOAD COMB 1012 1.2DL + 1.2LL + 1.2WL2  
103. 1 1.2 2 1.2 3 1.2 12 1.2  
104. LOAD COMB 1112 1.4DL + 1.4WL2  
105. 1 1.4 2 1.4 12 1.4  
106. LOAD COMB 1212 1.0DL + 1.4WL2  
107. 1 1.0 2 1.0 12 1.4  
108. LOAD COMB 1013 1.2DL + 1.2LL + 1.2WL3  
109. 1 1.2 2 1.2 3 1.2 13 1.2  
110. LOAD COMB 1113 1.4DL + 1.4WL3  
111. 1 1.4 2 1.4 13 1.4  
112. LOAD COMB 1213 1.0DL + 1.4WL3  
113. 1 1.0 2 1.0 13 1.4  
114. LOAD COMB 1014 1.2DL + 1.2LL + 1.2WL4  
115. 1 1.2 2 1.2 3 1.2 14 1.2  
116. LOAD COMB 1114 1.4DL + 1.4WL4  
117. 1 1.4 2 1.4 14 1.4  
118. LOAD COMB 1214 1.0DL + 1.4WL4  
119. 1 1.0 2 1.0 14 1.4  
120. LOAD COMB 1015 1.2DL + 1.2LL + 1.2WL5  
121. 1 1.2 2 1.2 3 1.2 15 1.2  
122. LOAD COMB 1115 1.4DL + 1.4WL5  
123. 1 1.4 2 1.4 15 1.4  
124. LOAD COMB 1215 1.0DL + 1.4WL5  
125. 1 1.0 2 1.0 15 1.4  
126. LOAD COMB 1016 1.2DL + 1.2LL + 1.2WL6  
127. 1 1.2 2 1.2 3 1.2 16 1.2  
128. LOAD COMB 1116 1.4DL + 1.4WL6  
129. 1 1.4 2 1.4 16 1.4  
130. LOAD COMB 1216 1.0DL + 1.4WL6  
131. 1 1.0 2 1.0 16 1.4  
132. LOAD COMB 1021 1.2DL + 0.5LL + 1.0EQ  
133. 1 1.2 2 1.2 3 0.5 21 1.0  
134. LOAD COMB 1121 1.2DL + 0.5LL - 1.0EQ  
135. 1 1.2 2 1.2 3 0.5 21 -1.0  
136. LOAD COMB 1221 0.9DL + 1.0EQ  
137. 1 0.9 2 0.9 21 1.0  
138. LOAD COMB 1321 0.9DL - 1.0EQ  
139. 1 0.9 2 0.9 21 -1.0  
140. LOAD COMB 2001 1.0DL + 1.0LL  
141. 1 1.0 2 1.0 3 1.0  
142. LOAD COMB 2002 1.0DL + 0.5LL  
143. 1 1.0 2 1.0 3 0.5  
144. LOAD COMB 2011 DL + LL + WL1  
145. 1 1.0 2 1.0 3 1.0 11 1.0  
146. LOAD COMB 2012 DL + LL + WL2  
147. 1 1.0 2 1.0 3 1.0 12 1.0  
148. LOAD COMB 2013 DL + LL + WL3  
149. 1 1.0 2 1.0 3 1.0 13 1.0  
150. LOAD COMB 2014 DL + LL + WL4  
151. 1 1.0 2 1.0 3 1.0 14 1.0  
152. LOAD COMB 2015 DL + LL + WL5

FRAME 1

-- PAGE NO. 4

153. 1 1.0 2 1.0 3 1.0 15 1.0  
154. LOAD COMB 2016 DL + LL + WL6  
155. 1 1.0 2 1.0 3 1.0 16 1.0  
156. PERFORM ANALYSIS

## P R O B L E M   S T A T I S T I C S

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NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =     9/     8/     2  
ORIGINAL/FINAL BAND-WIDTH=     7/     1/     6 DOF  
TOTAL PRIMARY LOAD CASES =   11, TOTAL DEGREES OF FREEDOM =     23  
SIZE OF STIFFNESS MATRIX =     1 DOUBLE   KILO-WORDS  
REQRD/AVAIL. DISK SPACE =     12.0/     919.3 MB

157. PRINT MEMBER FORCES